

Input Parameters (Global)		
	Metric	Imperial
Brake Bias (Front)	0.548	0.548
Static Loaded Radius	199.9	7.87
Pedal Force	471.5	106.0
Pedal Ratio	3.40	3.40

Input Parameters (Front)		
Effective Rotor Radius	72.50	2.85
MC Bore Diameter	15.88	0.6250
Pad Mu	0.50	0.50
Piston Diameter	34.00	1.34
Piston Quantity	4.00	4.00

Input Parameters (Rear)		
Effective Rotor Radius	76.50	3.01
MC Bore Diameter	15.88	0.6250
Pad Mu	0.50	0.50
Piston Diameter	24.00	0.94
Piston Quantity	4.00	4.00

Outputs (Per Axle)		
Caliper Piston Area (Front)	3631.7	5.629
Caliper Piston Area (Rear)	1809.6	2.805
MC Piston Area (Front)	197.9	0.307
MC Piston Area (Rear)	197.9	0.307
Longitudinal Force (Front)	2923.1	657.1
Longitudinal Force (Rear)	1267.6	285.0
Pedal Force Gradient	300.3	67.5

Ultimate Pressure		
Max Pedal Force	600.0	134.9
Line Pressure (Front)	5.6	819.2
Line Pressure (Rear)	4.7	675.7

Input Parameters - Expected Grip L	
	Metric
Vehicle + Driver Mass	2668.9
Wheel Base	1549.4
Decel (g)	1.57
CG Height	282.4
CG X	774.7
Downforce	330.0
CoP X	774.7

Static Normal Loads (Per Axle)	
Front	1499.5
Rear	1499.5

Front-Rear Load Transfer (Per Axle)	
Pitch Moment	1183521
Load Transfer	763.9
Normal Load (Front)	2263.3
Normal Load (Rear)	735.6

Longitudinal Capability (Per Axle)	
Mu Scaling Factor	0.60
Longitudinal Grip (Front)	2914.3
Longitudinal Grip (Rear)	1266.2
Decel (g)	1.5

% To Locking @ Input Force	
Front	100.3
Rear	100.1
Error (Front Bias)	0.19

imits
Imperial
600.0
61.0
1.57
11.1
30.5
74.2
30.5

)
337.1
337.1

le)
10475.0
171.7
508.8
165.4

e)
0.60
655.2
284.7
7

0%
.1%
0%

Rotor Dimensions (Front)		
Effective Rotor Radius	72.50	2.85
Pad Radial Width	33.1	1.30
Button Diameter	12.70	0.5
Button-Pad Clearance	3	0.12
Pad-OD Clearance	1.5	0.06
Outer Diameter	181.1	7.130
Inner Diameter	93.2	3.669

Rotor Dimensions (Rear)		
Effective Rotor Radius	76.50	3.01
Pad Radial Width	33.1	1.30
Button Diameter	12.70	0.5
Button-Pad Clearance	3	0.12
Pad-OD Clearance	1.5	0.06
Outer Diameter	189.1	7.445
Inner Diameter	101.2	3.984

INPUT

DRIVEN

LABEL

OUTPUT

Must be > 127 b/c diff

Input Parameters (Global)		
	Metric	Imperial
Bias Bar (Front)	0.417	0.417
Static Loaded Radius	199.9	7.87
Pedal Force	360.3	81.0
Pedal Ratio	3.50	3.50

Input Parameters (Front)		
Effective Rotor Radius	72.00	2.83
MC Bore Diameter	15.88	0.6250
Pad Mu	0.50	0.50
Piston Diameter	34.00	1.34
Piston Quantity	4.00	4.00

Input Parameters (Rear)		
Effective Rotor Radius	80.00	3.15
MC Bore Diameter	15.88	0.6250
Pad Mu	0.50	0.50
Piston Diameter	34.00	1.34
Piston Quantity	2.00	2.00

Outputs (Per Axle)		
Caliper Piston Area (Front)	3631.7	5.629
Caliper Piston Area (Rear)	1815.8	2.815
MC Piston Area (Front)	197.9	0.307
MC Piston Area (Rear)	197.9	0.307
Longitudinal Force (Front)	1737.6	390.6
Longitudinal Force (Rear)	1349.6	303.4
Pedal Force Gradient	311.5	70.0

Ultimate Pressure		
Max Pedal Force	360.0	80.9
Line Pressure (Front)	2.7	385.0
Line Pressure (Rear)	3.7	538.3

Input Parameters - Expected Grip Limits		
	Metric	Imperial
Vehicle + Driver Mass	2668.9	600.0
Wheel Base	1549.4	61.0
Decel (g)	2.06	2.06
CG Height	282.4	11.1
CG X	774.7	30.5
Downforce	1600.0	359.7
CoP X	774.7	30.5

Static Normal Loads (Per Axle)		
Front	2134.5	479.8
Rear	2134.5	479.8

Front-Rear Load Transfer (Per Axle)		
Pitch Moment	1552900	13744.3
Load Transfer	1002.3	225.3
Normal Load (Front)	3136.7	705.2
Normal Load (Rear)	1132.2	254.5

Longitudinal Capability (Per Axle)		
Mu Scaling Factor	0.65	0.65
Longitudinal Grip (Front)	3533.0	794.2
Longitudinal Grip (Rear)	1973.2	443.6
Decel (g)	2.06	

% To Locking @ Input Force		
Front	49.18%	
Rear	68.40%	
Error (Front Bias)	-19.22%	

Decel (g)	Nominal Bias		Most Forward Bias		Most Rearward Bias	
	1.16	-4%		90%		-6%
	1.26	-7%		85%		-9%
	1.36	-10%		79%		-11%
	1.46	-12%		74%		-14%
	1.56	-15%		69%		-16%
	1.66	-18%		64%		-19%
	1.76	-20%		59%		-22%

1.86	-23%	53%	-24%
1.96	-25%	48%	-27%
2.06	-19%	43%	-29%
2.16	-31%	38%	-32%
2.26	-33%	32%	-34%
2.36	-36%	27%	-37%
2.46	-39%	22%	-40%
2.56	-41%	16%	-42%
2.66	-44%	11%	-45%

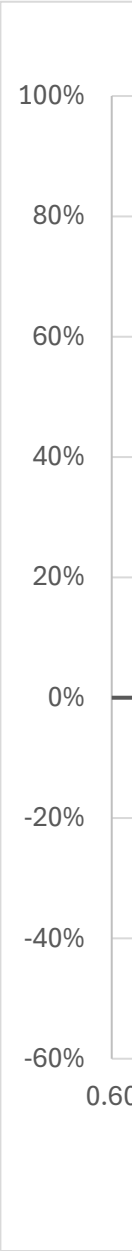
Rotor Dimensions (Front)		
Effective Rotor Radius	72.00	2.83
Pad Radial Width	33.1	1.30
Button Diameter	12.70	0.5
Button-Pad Clearance	3	0.12
Pad-OD Clearance	1.5	0.06
Outer Diameter	180.1	7.091
Inner Diameter	92.2	3.630

Rotor Dimensions (Rear)		
Effective Rotor Radius	80.00	3.15
Pad Radial Width	33.1	1.30
Button Diameter	12.70	0.5
Button-Pad Clearance	4	0.16
Pad-OD Clearance	1.5	0.06
Outer Diameter	196.1	7.720
Inner Diameter	106.2	4.181

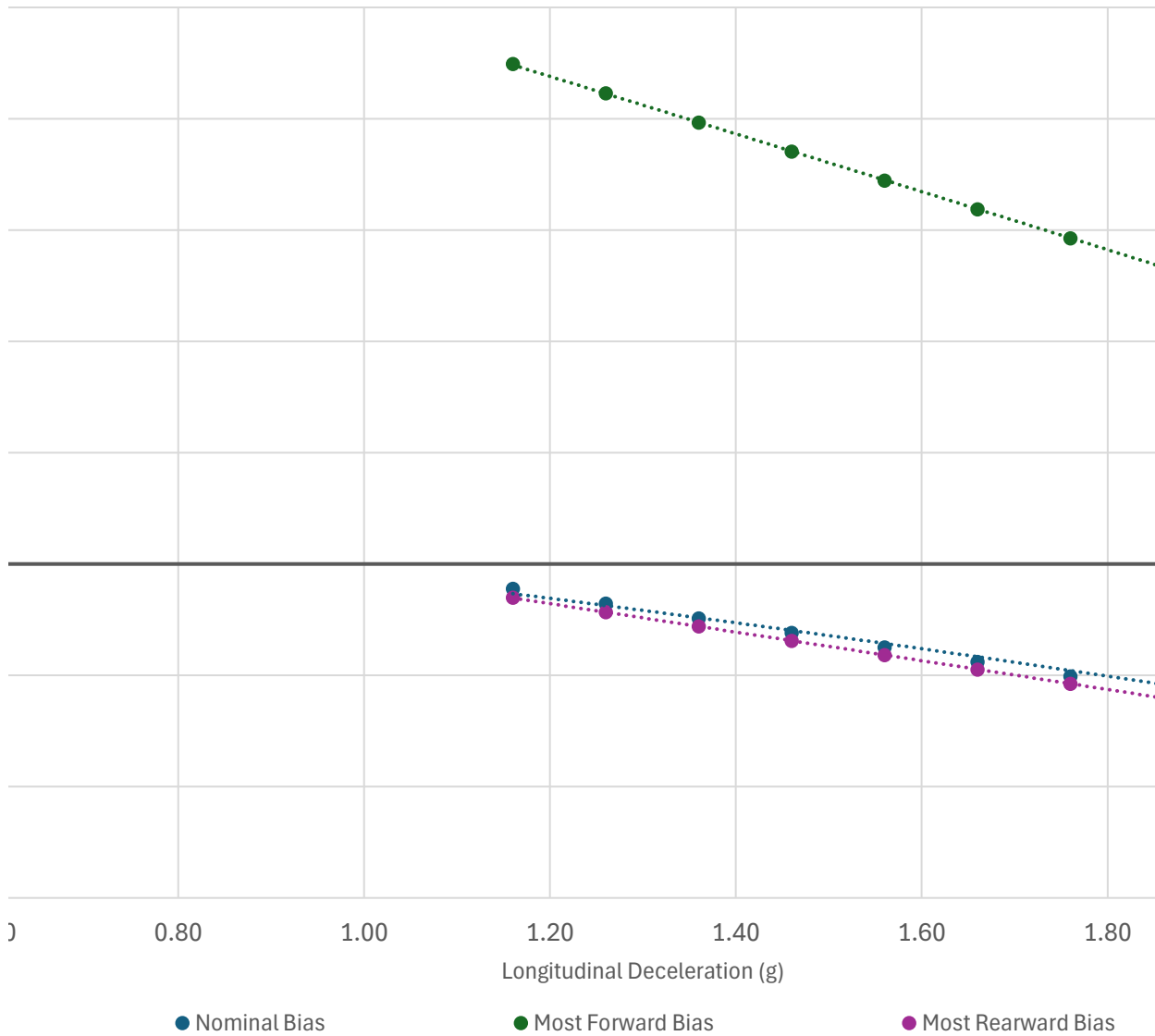
How to use this sheet:
1. Fill out vehicle inputs in G4:H10
2. Fill out brakes inputs in C4:D21
3. Change brakes inputs until blue cells meet target values
4. Reference J3:L18 for driven rotor dimensions

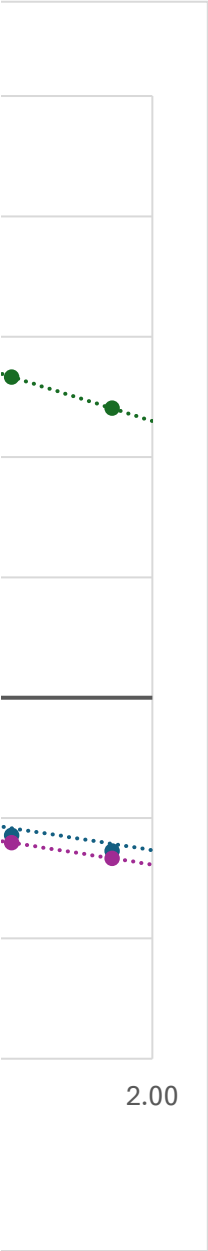
Targets:
0% Locking Bias at max decel
60 < Pedal Force Gradient < 70 lbf/g
Ultimate Pressure < Caliper Burst Ratings

INPUT
DRIVEN
LABEL
OUTPUT



Front Locking Error (%) vs Deceleration (g)





Input Parameters (Global)		
	Metric	Imperial
Brake Bias (Front)	0.562	0.562
Static Loaded Radius	199.9	7.87
Pedal Force	444.8	100.0
Pedal Ratio	3.50	3.50

Input Parameters (Front)		
Effective Rotor Radius	72.00	2.83
MC Bore Diameter	15.88	0.6250
Pad Mu	0.50	0.50
Piston Diameter	34.00	1.34
Piston Quantity	4.00	4.00

Input Parameters (Rear)		
Effective Rotor Radius	80.00	3.15
MC Bore Diameter	15.88	0.6250
Pad Mu	0.50	0.50
Piston Diameter	34.00	1.34
Piston Quantity	2.00	2.00

Outputs (Per Axle)		
Caliper Piston Area (Front)	3631.7	5.629
Caliper Piston Area (Rear)	1815.8	2.815
MC Piston Area (Front)	197.9	0.307
MC Piston Area (Rear)	197.9	0.307
Longitudinal Force (Front)	2891.2	650.0
Longitudinal Force (Rear)	1251.8	281.4
Pedal Force Gradient	286.6	64.4

Ultimate Pressure (@ 450lbs)		
Max Pedal Force	2000.0	449.6
Line Pressure (Front)	19.9	2882.7
Line Pressure (Rear)	15.5	2246.7

Input Parameters - Expected Grip L	
	Metric
Vehicle + Driver Mass	2668.9
Wheel Base	1549.4
Decel (g)	1.57
CG Height	282.4
CG X	774.7
Downforce	330.0
CoP X	774.7

Static Normal Loads (Per Axle)	
Front	1499.5
Rear	1499.5

Front-Rear Load Transfer (Per Axle)	
Pitch Moment	1183521
Load Transfer	763.9
Normal Load (Front)	2263.3
Normal Load (Rear)	735.6

Longitudinal Capability (Per Axle)	
Mu Scaling Factor	0.60
Longitudinal Grip (Front)	2914.3
Longitudinal Grip (Rear)	1266.2
Decel (g)	1.5

% To Locking @ Input Force	
Front	99.2
Rear	98.8
Error (Front Bias)	0.35

Caliper tangential force = tire long fc
3163.852179 711.26226
Newtons lbf

imits
Imperial
600.0
61.0
1.57
11.1
30.5
74.2
30.5

)
337.1
337.1

le)
10475.0
171.7
508.8
165.4

e)
0.60
655.2
284.7
7

1%
7%
5%

Rotor Dimensions (Front)		
Effective Rotor Radius	72.00	2.83
Pad Radial Width	33.1	1.30
Button Diameter	12.70	0.5
Button-Pad Clearance	3	0.12
Pad-OD Clearance	1.5	0.06
Outer Diameter	180.1	7.091
Inner Diameter	92.2	3.630

Rotor Dimensions (Rear)		
Effective Rotor Radius	80.00	3.15
Pad Radial Width	33.1	1.30
Button Diameter	12.70	0.5
Button-Pad Clearance	3	0.12
Pad-OD Clearance	0.5	0.02
Outer Diameter	194.1	7.642
Inner Diameter	108.2	4.260

How to use this sheet:
1. Fill out vehicle inputs in G4:H10
2. Fill out brakes inputs in C4:D21
3. Change brakes inputs until blue cells meet target values
4. Reference J3:L18 for driven rotor dimensions

Targets:
0% Locking Bias (pref @ 50% brake bias)
60 < Pedal Force Gradient < 70 lbf/g
Ultimate Pressure < Caliper Burst Ratings

INPUT

DRIVEN

LABEL

OUTPUT

orce * tire radius / rotor radius

Input Parameters (Global)		
	Metric	Imperial
Brake Bias (Front)	0.583	0.583
Static Loaded Radius	199.9	7.87
Pedal Force	431.5	97.0
Pedal Ratio	3.50	3.50

Input Parameters (Front)		
Effective Rotor Radius	72.00	2.83
MC Bore Diameter	15.88	0.6250
Pad Mu	0.50	0.50
Piston Diameter	34.00	1.34
Piston Quantity	4.00	4.00

Input Parameters (Rear)		
Effective Rotor Radius	80.00	3.15
MC Bore Diameter	15.88	0.6250
Pad Mu	0.50	0.50
Piston Diameter	34.00	1.34
Piston Quantity	2.00	2.00

Outputs (Per Axle)		
Caliper Piston Area (Front)	3631.7	5.629
Caliper Piston Area (Rear)	1815.8	2.815
MC Piston Area (Front)	197.9	0.307
MC Piston Area (Rear)	197.9	0.307
Longitudinal Force (Front)	2909.2	654.0
Longitudinal Force (Rear)	1156.0	259.9
Pedal Force Gradient	283.3	63.7
Ultimate Pressure (@ 450lbs)		
Max Pedal Force	2000.0	449.6
Line Pressure (Front)	20.6	2990.4
Line Pressure (Rear)	14.7	2138.9
Front Wheel Torque Required (Per Axle)	581.6	428.9
Rear Wheel Torque Required (Per Axle)	231.1	170.4
Line Pressure @ Input Force (Front)	4.4	645.1
Line Pressure @ Input Force (Rear)	3.2	461.5

Input Parameters - Expected Grip Limits		
	Metric	Imperial
Vehicle + Driver Mass	2668.9	600.0
Wheel Base	1549.4	61.0
Decel (g)	1.57	1.57
CG Height	282.4	11.1
CG X	774.7	30.5
Downforce	330.0	74.2
CoP X	774.7	30.5

Static Normal Loads (Per Axle)		
Front	1499.5	337.1
Rear	1499.5	337.1

Front-Rear Load Transfer (Per Axle)		
Pitch Moment	1183521	10475.0
Load Transfer	763.9	171.7
Normal Load (Front)	2263.3	508.8
Normal Load (Rear)	735.6	165.4

Longitudinal Capability (Per Axle)		
Mu Scaling Factor	0.60	0.60
Longitudinal Grip (Front)	2914.3	655.2
Longitudinal Grip (Rear)	1266.2	284.7
Decel (g)	1.57	

% To Locking @ Input Force		
Front	99.83%	
Rear	91.30%	
Error (Front Bias)	9.34%	

Rotor Dimensions (Front)		
Effective Rotor Radius	72.00	2.83
Pad Radial Width	33.1	1.30
Button Diameter	12.70	0.5
Button-Pad Clearance	3	0.12
Pad-OD Clearance	1.5	0.06
Outer Diameter	180.1	7.091
Inner Diameter	92.2	3.630

Rotor Dimensions (Rear)		
Effective Rotor Radius	80.00	3.15
Pad Radial Width	33.1	1.30
Button Diameter	12.70	0.5
Button-Pad Clearance	3	0.12
Pad-OD Clearance	0.5	0.02
Outer Diameter	194.1	7.642
Inner Diameter	108.2	4.260

How to use this sheet:	
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4. Reference J3:L18 for driven rotor dimensions	

Targets:	
0% Locking Bias (pref @ 50% brake bias)	
60 < Pedal Force Gradient < 70 lbf/g	
Ultimate Pressure < Caliper Burst Ratings	

Caliper tangential force = tire long force * tire radius / rotor radius

3163.852179 711.26226

Newtons

lbf

INPUT

DRIVEN

LABEL

OUTPUT